

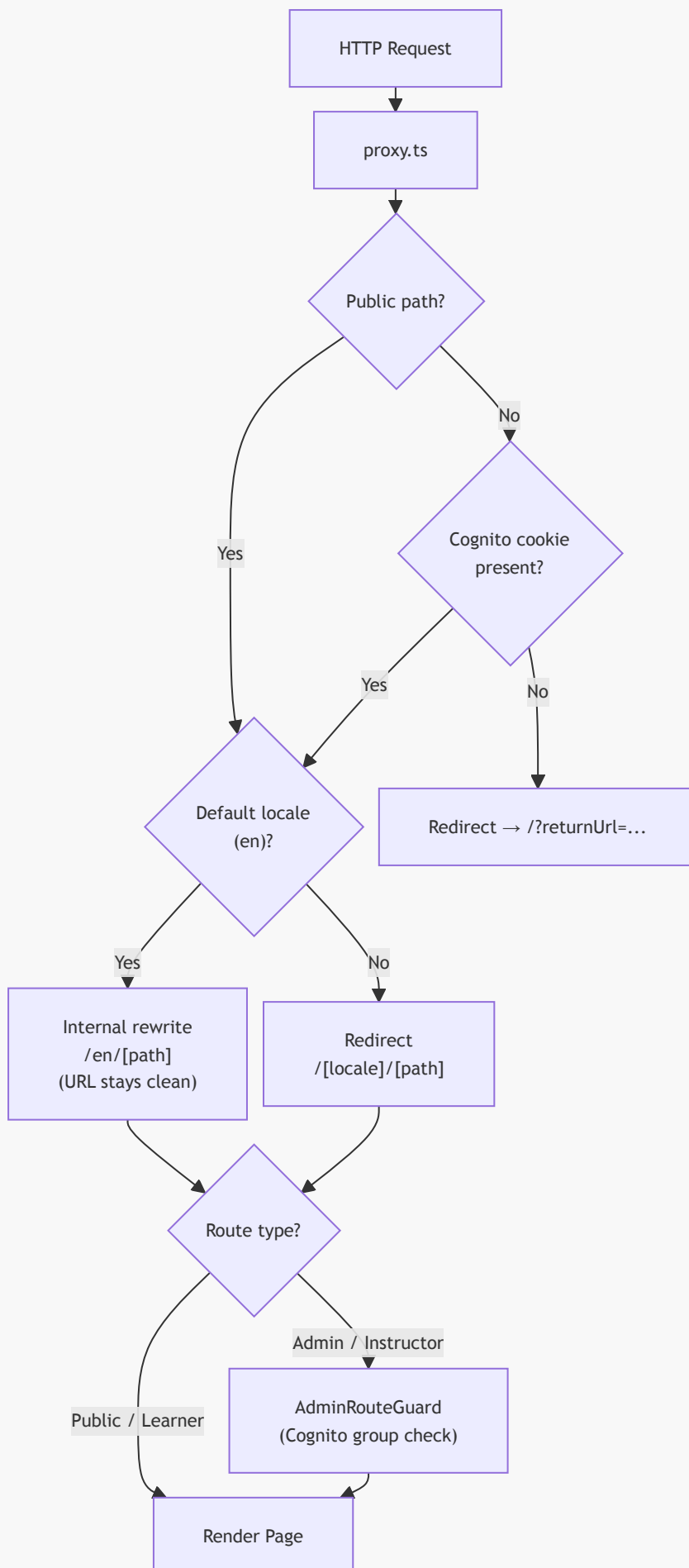
Main Feature List — Homework Supply

Source-validated against actual code in routes, contexts, schema, Lambda handlers, and package dependencies.

Last updated: 2026-07-14 (re-validated via exhaustive code exploration)

1. User Roles & Auth

- Cognito-based auth with groups: **Admins, Instructors, Learners**, plus dynamic per-section and peer-review groups defined via `groupDefinedIn` fields in the schema
- Route guards on admin/instructor routes
(`app/[locale]/admin/_components/AdminRouteGuard.tsx`, `proxy.ts`)
- Owner + group + authenticated-read auth blends per model
- Role-constrained custom mutations (instructor-only submission URL retrieval, publish/index actions)
- Centralized auth/session context with token/group extraction and lifecycle handling
(`src/context/authContext.jsx`)

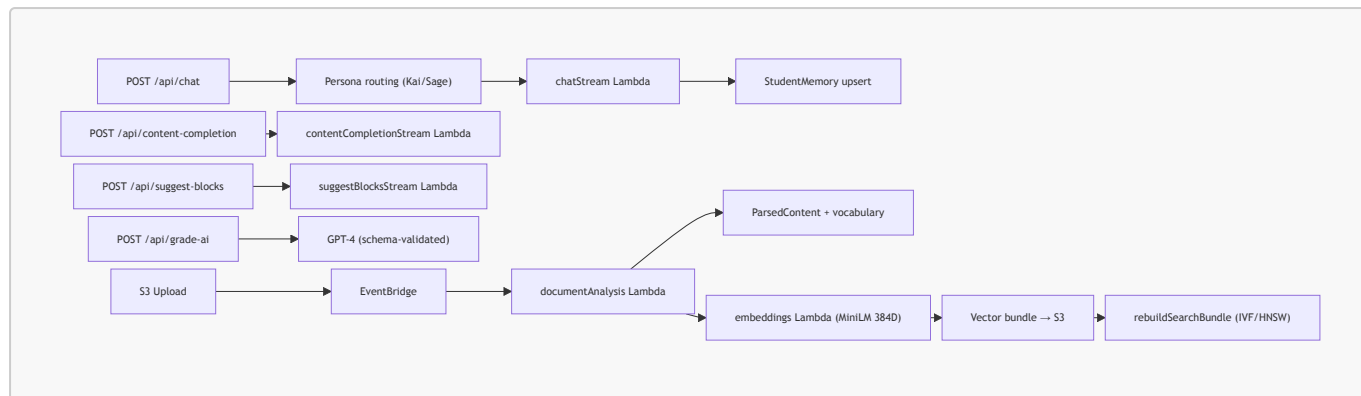


2. Core Learning & Content

- **Unit authoring** — create, edit, and publish learning modules using the rich Lexical editor
 - **Learner dashboard** — assignments, grade state, sections, and recommendations
 - **Workbook execution** — timer-gated starts, rubric grading, completion tracking, and peer-review launch points
 - **Sections / classes** — create and manage classes, student enrollments, assignment mapping, and gradebook
 - **Vocabulary & question authoring** — word/definition/audio pipelines and question banks
 - **Instructor grade review** — multi-attempt context and moderation metadata
 - **Student profiles** — progression, streak calendar, badge visibility controls
 - **Peer review rooms** — AI-assisted `@mention` handling and review summary generation
 - **Practice drills** — AI-generated drill content per unit
 - **Leaderboard** — platform-wide and section-scoped student rankings
-

3. AI & ML Features

- **Streaming AI chat** — tool use, persona routing, token accounting, chat memory summarization (`app/api/chat/route.ts`, `app/api/_shared/botPersonas.ts`)
- **Editor AI autocomplete** — inline content completion while authoring (`app/api/content-completion/route.ts`)
- **AI block suggestions** — structured tool outputs for inserting Lexical editor blocks (`app/api/suggest-blocks/route.ts`)
- **AI grading** — prompt-hardened, schema-validated, PII-filtered grading endpoint (`app/api/grade-ai/route.ts`)
- **Document analysis** — PDF text extraction and media approval producing parsed vocabulary and educational artifacts; also triggered automatically via EventBridge on S3 upload (`amplify/functions/documentAnalysis/`)
- **Embedding generation + vector search** — `Xenova/all-MiniLM-L6-v2` (384D, offline-compatible); linear/IVF/HNSW architecture with client-side search bundles (`app/actions/embeddings.ts`, `amplify/functions/embeddings/`, `docs/SEARCH_ARCHITECTURE.md`)
- **Content moderation** — text, image, and audio moderation pipeline (`amplify/functions/moderation/`, `app/actions/moderate.ts`)
- **Peer review AI** — handles `@mentions` and generates review summaries (`amplify/functions/peerReviewAI/`)
- **Practice drill generation** — AI creates drill content from unit/section context (`amplify/functions/generatePracticeDrill/`)
- **Skill tree generation** — AI evaluates learner skills per unit and advances progression
- **Student memory** — per-student contextual memory upsert, profile rebuild, and unit-grade-based updates

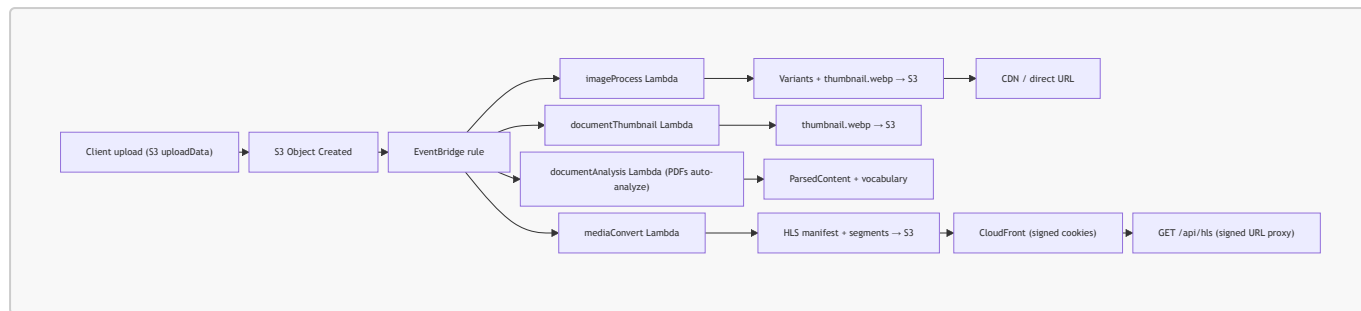


4. Editor Features

- **Lexical rich text editor** with SSR skeleton and workbook rendering mode
([src/components/Editor3/](#))
- **Graded block types:** [quiz](#), [answer](#), [meaning-association](#), [custom-answer](#), [custom-ai](#)
- **Media blocks:** image, PDF viewer, playlist, table, Excalidraw diagram, file metadata
- **Assignment configuration** — due dates, timer, attempt limits
([src/components/Editor3/components/AssignmentConfiguration.jsx](#))
- **Grade history viewer** embedded within editor context
([src/components/Editor3/components/GradeHistory.jsx](#))
- **Chat-assisted block insertion** — preview and insert AI-suggested blocks from the sidebar
([src/components/ChatSidebar/BlockInsertPreview.jsx](#))
- **Markdown support** via Lexical markdown plugin
- **Vocabulary autocomplete** plugin for inserting dictionary words inline

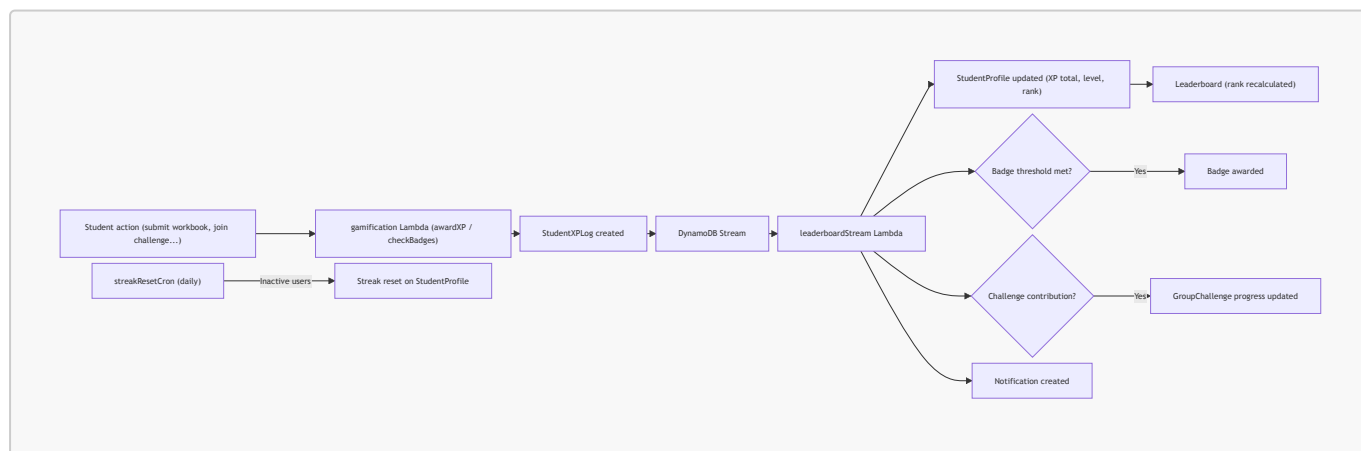
5. File & Media Management

- **S3 file management** with public / protected / private storage levels
([src/context/fileContext.jsx](#))
- **Image processing** — Lambda generates multiple image variants and PDF first-page thumbnails
([amplify/functions/imageProcess/](#))
- **Office / document thumbnails** — LibreOffice Lambda layer for document-to-image conversion
([amplify/functions/documentThumbnail/](#))
- **Video transcoding** — AWS MediaConvert pipeline producing HLS adaptive bitrate streams
([amplify/functions/mediaConvert/](#))
- **HLS proxy** — signed URL rewriting for HLS segment delivery ([app/api/hls/route.ts](#))
- **CloudFront OAC** — signed cookie media delivery for published units
- **Soft delete / restore / permanent archive** — full recycle bin lifecycle
([amplify/functions/recycleBin/](#), [app/\[locale\]/recycle-bin/](#))
- **Admin unarchive** — administrators can restore any archived record
([app/\[locale\]/admin/archives/](#))



6. Gamification

- **XP system** — award, log, history view, toast notifications, level-up celebration
([src/context/xpContext.tsx](#), [app/\[locale\]/xp-history/](#))
- **Streaks** — daily streak tracking with scheduled cron reset
([amplify/functions/streakResetCron/](#))
- **Badges** — standard badges, anti-badges, custom badges with visual editor
([src/components/Gamification/BadgeEditor.tsx](#))
- **Easter eggs** — secret links, hidden triggers, discoverable events
([src/components/Gamification/EasterEggLayer.tsx](#))
- **Skill trees** — AI-generated skill trees with visual progression UI
([src/components/Gamification/SkillTree.tsx](#), [src/context/skillTreeContext.tsx](#))
- **Squads** — create/join squads, squad posts, squad leaderboard, membership management
([app/\[locale\]/squads/](#), [src/components/Gamification/SquadLeaderboard.tsx](#))
- **Group challenges / boss battles** — campaign timelines and contribution tracking
([src/components/Gamification/BossBattleCard.tsx](#),
[src/components/Gamification/CampaignTimeline.tsx](#))
- **Content locks** — gamification-gated content cards
([src/components/Gamification/ContentLockCard.tsx](#))
- **Personal bests** — tracked per-student per-unit
- **Per-section and platform gamification config** — overrideable settings at section and admin level



7. Analytics

- **Event ingestion API** — auth + geo enrichment from CloudFront headers
([app/api/analytics/route.ts](#))

- **Web vitals ingestion** — payload/rate-controlled Core Web Vitals collection (`app/api/vitals/route.ts`)
 - **Kinesis-backed aggregation** — Lambda aggregates events with scope-based rollups: platform / section / unit / squad / page / geo (`amplify/functions/analyticsAggregator/`)
 - **Admin analytics dashboard** with dimension filters (`app/[locale]/admin/analytics/`)
-

8. Admin & Governance

- **Moderation dashboard** — flagged-item review and resolution workflows (`app/[locale]/admin/moderation/`)
 - **Vocabulary maintenance** — batch delete and diagnostics (`app/[locale]/admin/words/`)
 - **Platform settings** — AI feature toggles and gamification controls (`app/[locale]/admin/settings/`)
 - **Archive management** — list, view, and restore archived records (`app/[locale]/admin/archives/`)
 - **Instructor AI override** — per-section AI settings (`app/[locale]/section/[id]/settings/ai/`)
 - **Admin user management** — Cognito admin operations via management scripts
-

9. Collaboration & Real-Time

- **Real-time data subscriptions** across all core models (units, grades, sections, words, files, chat, notifications) via Amplify `observeQuery`
 - **Yjs collaborative infrastructure** — WebSocket sync Lambda, Yjs hooks for shared editor state (`amplify/functions/yjsSync/, src/yjs/`)
 - **Notifications** — real-time feed with unseen count badge (`src/context/notificationContext.tsx, app/[locale]/profile/notifications/`)
 - **Workbook comments** — inline commenting on student submissions
-

10. Offline & PWA

- **Service worker** (Serwist) with offline caching strategy (`src/sw.ts`)
 - **Offline workbook page** — degraded-mode workbook experience when network is unavailable (`app/[locale]/workbook-offline/`)
 - **PWA manifest** — installable app with icons and theme configuration (`public/manifest.json`)
-

11. Search

- **N-gram index rebuild** — Lambda-based n-gram search index for vocabulary and content (`amplify/functions/rebuildNgramIndex/`)
- **Vector search bundles** — linear/IVF/HNSW client-side search (HNSW for ≥ 5000 vectors) with Lambda-powered bundle rebuild (`amplify/functions/rebuildSearchBundle/, src/utils/searchBundles.ts`)
- **Global search context** — unified search across units, words, and files (`src/context/searchContext.tsx`)

12. Internationalization

- **Multi-locale routing** via `next-intl` — locale-scoped pages under `app/[locale]/`; 6 supported locales: `de`, `en`, `es`, `fr`, `ja`, `zh`
- **Clean default-locale URLs** — `/units` rewrites internally to `/en/units` (no prefix in URL)
- **Non-default locale redirects** — e.g. `/es/units` for Spanish
- Translation JSON files in `public/locales/`

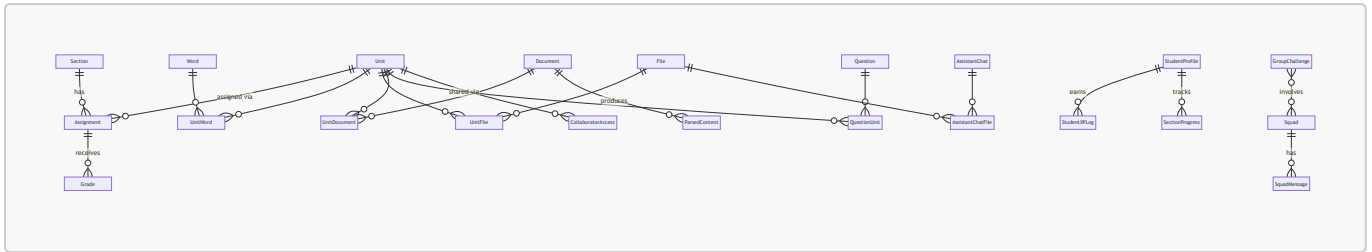
13. Technical Infrastructure

- **Next.js App Router** — server prefetch + client hydration; no legacy `pages/` directory
- **AWS Amplify Gen 2** — DynamoDB, AppSync GraphQL, Cognito, S3, Lambda
- **Optimistic locking** — `_version` fields on all models for concurrent update safety
- **Scheduled Lambda crons** — streak reset, notification dispatch
- **Vercel AI SDK** — streaming chat with `useChat` hook and tool call support
- **Material UI** component library
- **Storybook** component development with mocked AWS services and mock data in `.storybook/__mocks__/`
- **Cypress E2E** tests, **Vitest** unit/integration tests, **Playwright** browser tests
- **TypeScript migration in progress** — `allowJs: true`, new files as `.tsx`
- **Agent permissions system** — file-operation gating with audit logs (`scripts/agent-cli.ts`, `.github/agent-permissions.json`)

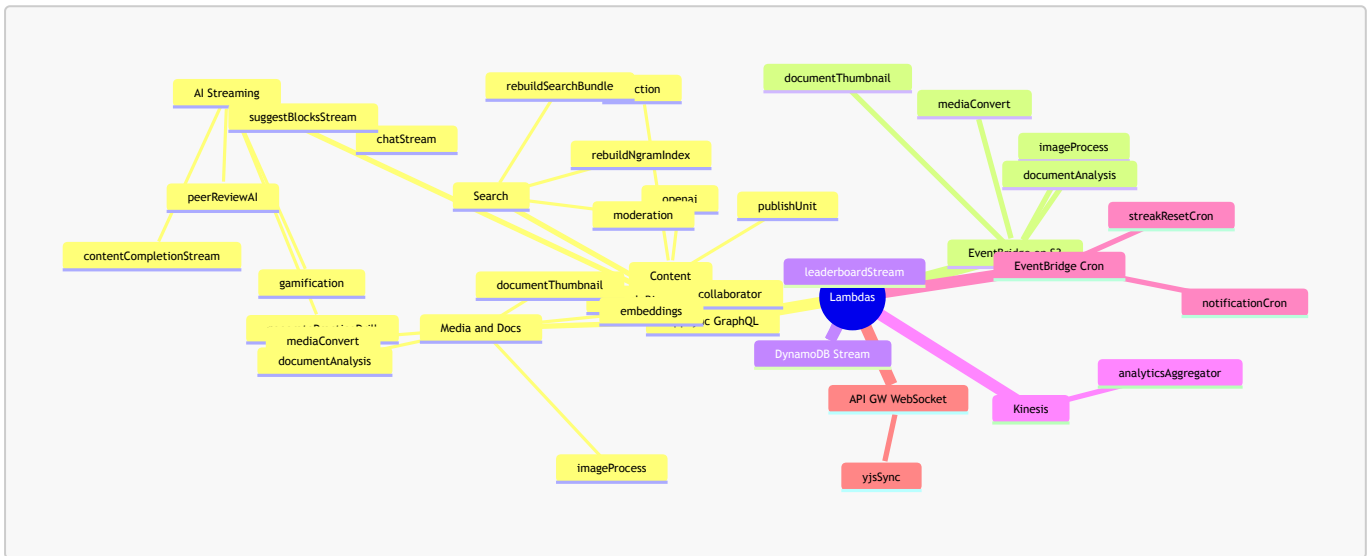
14. Data Models

38 primary models plus join tables:

Category	Models
Content	Unit, Assignment, Grade, Section, Question, Word, File, Document, ParsedContent
AI / Chat	AssistantChat, AgentJob, AIFeedback, StudentMemory
Users	Settings, StudentProfile, StudentXPLog, SectionProgress
Gamification	Badge, Skill, Squad, GroupChallenge, EasterEgg, PracticeSession, SquadMessage
Platform	PlatformSettings, Notification, WorkbookComment, HomeworkRoom, AnalyticsSummary
Join tables	UnitFile, UnitWord, QuestionUnit, UnitDocument, QuestionFile, WordFile, QuestionWord, DocumentWord, DocumentQuestion, AssistantChatFile, CollaboratorAccess



23 GraphQL resolver Lambda functions + 1 WebSocket server under `amplify/functions/`. Event-driven and WebSocket triggers; AppSync resolver functions are detailed in the table below.



Function	Purpose
<code>openai</code>	GPT-4, Whisper, TTS, image analysis, and feedback summarization via OpenAI
<code>section</code>	Section lifecycle — create groups, enroll students, peer-review operations, submission URL signing, CDN cookie issuance
<code>embeddings</code>	Generate and store vector embeddings (Xenova/all-MiniLM-L6-v2)
<code>moderation</code>	Text, image, and audio content moderation
<code>documentAnalysis</code>	PDF extraction, <code>approveMedia</code> mutation, and EventBridge S3 auto-analysis trigger
<code>mediaConvert</code>	AWS MediaConvert → HLS transcoding
<code>imageProcess</code>	Image variant generation and PDF thumbnails
<code>documentThumbnail</code>	LibreOffice document thumbnail conversion
<code>gamification</code>	XP awards, badge checks, streak updates, leaderboard, memory, challenges, skills, battle-stakes
<code>generatePracticeDrill</code>	AI-generated practice drills with TTS audio

Function	Purpose
<code>peerReviewAI</code>	Peer review AI mentions and summary generation
<code>chatStream</code>	Streaming AI chat handler
<code>contentCompletionStream</code>	Streaming editor autocomplete
<code>suggestBlocksStream</code>	Streaming AI block suggestions
<code>leaderboardStream</code>	DynamoDB stream consumer — updates StudentProfile/SectionProgress, badge checks, notifications, challenge progress
<code>analyticsAggregator</code>	Kinesis-backed event aggregation
<code>notificationCron</code>	Scheduled notification dispatch
<code>streakResetCron</code>	Scheduled daily streak resets
<code>publishUnit</code>	Unit publishing — S3 content write, path rewrites, section outline metadata updates, CloudFront signed cookies
<code>rebuildNgramIndex</code>	N-gram search index rebuild
<code>rebuildSearchBundle</code>	Linear/IVF/HNSW vector search bundle rebuild
<code>recycleBin</code>	Soft delete, restore, permanent delete, unarchive, listArchives, getArchive
<code>collaborator</code>	Validates grant/revoke collaborator access (Cognito group + ownership checks)
<code>yjsSync</code>	Yjs WebSocket server for real-time collaborative editor sync (not a GraphQL resolver)

16. API Routes (Next.js)

Route	Purpose
<code>POST /api/chat</code>	Streaming AI chat with tool use
<code>POST /api/content-completion</code>	Streaming editor content completion
<code>POST /api/suggest-blocks</code>	AI block suggestions for editor
<code>POST /api/grade-ai</code>	AI-assisted grading
<code>GET /api/hls</code>	HLS proxy with signed URL rewriting
<code>POST /api/analytics</code>	Client event ingestion
<code>POST /api/vitals</code>	Web vitals ingestion

17. Page Routes

Route	Audience
/	Learner dashboard
/units	Unit library
/unit/[id]	Unit editor
/workbook/[id]	Workbook (student execution)
/sections	Section list
/section/[id]	Section detail and gradebook
/section/[id]/settings/ai	Section AI override (instructor)
/section/[id]/settings/gamification	Section gamification config (instructor)
/instructor/grade/[id]	Grade review (instructor)
/leaderboard	Platform leaderboard
/drill/[id]	Practice drill
/squads	Squad directory
/squad/[id]	Squad detail
/review/[id]	Peer review room
/xp-history	XP log
/settings	User settings
/profile	Own profile (current user)
/profile/[username]	Student public profile
/profile/notifications	Notification feed
/privacy	Privacy policy
/recycle-bin	Recycle bin
/offline	Offline fallback
/workbook-offline	Offline workbook fallback
/admin/analytics	Admin analytics dashboard
/admin/moderation	Admin moderation queue
/admin/settings	Admin platform settings
/admin/words	Admin vocabulary management
/admin/archives	Admin archive management